Bio model behaviour probes: FLORA T2, FLORA T1, PH163 (+ Molek from docs): report, 2026-11-05

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EXPLORATORY: not pre-registered, small n, nothing here is a benchmark number or a SOTA claim. Test cases and the expectations written BEFORE the runs: `TEST-CASES.md`*. Raw results:* `results/*.json`*. Screenshots:* `screenshots/`*. Scripts:* `run_*.py` *(plus* `fetch_probe_sequences.py`*). Probe proteins: UniProt (CC BY 4.0), accession + sha256 in* `probes/proteins.json`*. BIO-BENCH-001 TEST labels were not used; the six FLORA UI examples were used only as known-answer controls.*

Setup honesty: FLORA T2 = production service on :8767; PH163 = a THROWAWAY QA instance (:18769, own secret, user qa-probe, allowlist v0.1 exact) because the production PH163 service on :8768 was found DOWN when I opened its page (not restarted); FLORA T1 = the T1 service source recovered from an old stash commit and run from a scratch folder on :8766 (all file pins matched: trainer, adapter, card v5 README, precision JSON), so its batch endpoint returning 404 may be a version difference, not a model bug.

1 FLORA T2 (ESM-2 650M frozen + mean-pool head)

Case	Result
T2-A panel (24 rice proteins whose class comes from UniProt keywords)	accuracy 18/24 = 75%: enzyme 8/8, storage 5/8, transport 5/8. Most misses look like keyword-vs-class problems rather than clear model errors (hypothesis, not checked): a vacuolar-processing ENZYME tagged storage, Rab5A (a GTPase) and eIF tagged transport, two "seed allergenic proteins" called other with margin 0.79/0.84.
T2-A non-plant controls	status "clear" on 4 of 9 (TP53 other, RNase A other, CA2 enzyme, insulin other), "ambiguous" on GFP, HBB, lysozyme, albumin, actin. RNase A (an enzyme) is called "other" with a clear margin; albumin (a transport protein) "other", ambiguous.
T2-B1 shuffles (24 shuffled sequences)	nearly all flip to "other"; some stay "clear" (e.g. margin 0.46-0.58 on a shuffled chitinase): "other" behaves like a sink class.
T2-B2 homopolymers	poly-Lys x100 -> structural with margin 0.98 "clear separation" (T2 and T1 both structural; T2 far

Case	Result
	more confident). poly-Gly -> other 0.61 “clear”. The service’s own status flag does not catch nonsense input.
T2-B3 point mutants	label unchanged in 8/8 for each of 6 proteins (48/48).
T2-B4 truncations	label holds to 50% in most cases; a transport protein kept its label even at 25%; others flip to other/enzyme at 25%.
T2-C input edges	lowercase, FASTA header, wrapped lines all parsed; X, U, `*` and length 1/5/1022 flagged out_of_distribution; length 1023, multi-record FASTA, junk characters and a wrong JSON field -> HTTP 422. **BUG: an empty sequence returns HTTP 500** (should be 422; T1 returns a clean 400).
T2-D attribution vs UniProt active sites	pooled 3/9 annotated active-site residues fall in the top 20% of attribution (chance ~1.8/9); per-protein permutation p-values 0.195 to 1.0, none significant. Underpowered and inconclusive, NOT evidence of localisation. One favourable example is shown in `screenshots/04_...` (rice DSP2 Q0DX67: the strongest attribution sits on the HCxxGxxR phosphatase motif around the annotated Cys143), but it is the best of the 8 (site at the 94th percentile), so treat it as an illustration, not a result.
T2-E point edit of the catalytic residue	Ala at the annotated catalytic site lowers P(enzyme) more than Ala at random positions in 8/8 proteins, and more than Ala at positions with the SAME wild-type residue (control run) in 8/8 (0-17% of matched positions below the site; matched n as low as 2 to 6 in three proteins). Effects are small (0.002 to 0.038 in probability). The pooled head seems to register single catalytic-residue changes even though 8-residue window masking could not localise them (hypothesis: the window is too coarse).

2 FLORA T1 vs T2 (39 proteins, identical inputs)

- Label agreement 79.5%. Accuracy on the 30 labelled items: T1 27/30 = 90% (transport 8/8, storage 6/8, enzyme 8/8, UI examples 5/6), T2 24/30 = 80% (transport 5/8, storage 5/8, enzyme 8/8, UI examples 6/6). CAUTION: T1 was trained on the same labelling scheme (UniProt keyword precedence) and on reviewed crop proteins, so most of these rice proteins are probably in-distribution for T1; this is NOT evidence that

the 150M model beats the 650M one. The T1 card itself reports macro-F1 0.73-0.75 on BIO-BENCH-001 T1 TEST and no measured gain over a mean-pooled frozen baseline (delta −0.049, CI includes 0).

- Documented T1 failure (germin-like called storage): reproduced by T1 (storage 0.93, shown with a "low reliability class: Storage calls were right 7 of 50 times (14%)" badge, `screenshots/06_...`); T2 returns enzyme at 0.455, ambiguous. The "expected" enzyme label for germin is my assumption.
- Confidence: median top score on in-scope rice proteins T1 0.978 vs T2 0.682; non-plant T1 0.80 vs T2 0.57; shuffled rice T1 0.65 vs T2 0.51. Both are uncalibrated; T1 is far more peaked.
- Shuffles: T1 often keeps the label (a shuffled vacuolar-processing enzyme stays "enzyme" at 0.99; shuffled glutelin becomes "other" at 0.92-0.97), i.e. T1 leans on composition; T2 sends almost every shuffle to "other".
- Edge handling: T1 gives a clean 400 on empty input where T2 returns 500; T1 raises no out-of-distribution flag for X residues or 1-5 residue inputs (low_reliability stays False), where T2 flags them; T1 `/classify_batch` returned 404 in this recovered service.

3 PH163 variant effect (QA gate instance)

Case	Result
PH-A familiarity (median p(wt))	ubiquitin 0.78, insulin 0.68, rice DSP2 0.54, lysozyme 0.53, HBB 0.44, RNase A 0.40, rice seed Q84UL5 0.22, TP53 0.14, GFP 0.06. The UI shows "Model does not recognise this protein. Treat every score as unreliable" for GFP (`screenshots/08_...`, scan 280 s on CPU) and a washed-out heatmap; ubiquitin gives a full-contrast heatmap in 22 s (`07_...`). TP53 is low (0.14) though a famous protein; probably its large disordered regions.
PH-B TP53 hotspots	R248Q 5th percentile (most disruptive 5% of its substitutions), R282W 10th, G245S 12th, R273H 18th, R175H 19th: all in the most-disruptive fifth, as expected.
PH-B sickle cell	HBB E7V (HbS, UniProt numbering) at the 63rd percentile and E7K (HbC) at the 76th, i.e. milder than most substitutions: the expected blind spot, because the HbS defect is polymerisation, not unfolding. A good "limits of a fold-level model" example.
PH-B AUROC, disease vs polymorphism	TP53 0.69 (CI 0.53-0.84) but with only 5 polymorphism-like negatives, HBB 0.47 (CI 0.24-0.72) with 7 positives. My labelling heuristic (text of UniProt natural variants) is crude, so these two numbers are NOT meaningful; a proper variant benchmark is the ProteinGym run (mean Spearman 0.446, identical to the official ESM-2 650M).

Case	Result
PH-D scoring and errors	`/score` equals the scan matrix to 3 decimals (6/6); wt mismatch, position 0 / past the end, non-standard residue, scan over 1022 and scan with X all give clean HTTP 400; lowercase accepted; a 1300-residue protein is windowed and flagged.
PH-F gated design (rice seed, password session)	wrong password 401; correct login 200; 4 proposals; every proposal the same length as the seed with the requested number of substitutions; `frozen` positions and `avoid` residues respected (0 violations); `max_mutations=1` respected. Joint rescore vs additive estimate: 6.72 vs 4.78 for the 3-substitution proposal (epistasis +1.94), 3.15 vs 3.27 for 2-substitution, equal for single substitutions (`screenshots/09_...`, `10_...` show the 3D check with ddG).
PH-F negative controls	all 7 refused with one refusal shape (HTTP 403): no token, bogus token, non-allowlisted protein (GFP), allowlisted seed with one residue edited, wrong purpose, lowercase seed, empty sequence. Usability note: a lowercase copy of an allowlisted seed is refused rather than normalised (exact mode is case-sensitive).
PH-C structure	seed structure and ddG table returned for the rice seed (precomputed); live folding needs the pod.

4 Molek v1 (documentation only; not run)

See `MOLEK-DOC-REVIEW.md`: nine mistakes and how each was caught (fragment-salvage 99.95% with raw validity 0%, FCD scale error, the unsourced 25x that is 4.49x, fabricated card numbers, inert conditioning, wrong benchmarks, serving bugs). The v1 weights exist locally (sha256 verified) but outside git.

5 Issues for the owners

1. FLORA T2: empty sequence -> HTTP 500 (should be 422). Owner: ml-engineer.
2. FLORA T2 / T1: the status flag can say "clear separation" on shuffled or homopolymer input; the "other" class absorbs nonsense. A length-and-composition sanity flag would help.
3. FLORA T1 service (this recovered version): no OOD flags for X or very short input; batch endpoint 404; the current live T1 service source is only in an old stash commit (9e87242), not on a branch.
4. PH163 production service (port 8768) was not running during this session (cause not investigated).
5. The T2 attribution is too coarse to localise active sites; a single-residue edit scan (as in T2-E) is a more sensitive, still exploratory, diagnostic.

6 Suggested things to share (Stan decides; all exploratory)

"The model's own warning works on GFP", the TP53 hotspot percentiles, the sickle-cell blind spot, the shuffled-sequence "sink" behaviour and the T1 over-confidence, the catalytic-residue edit effect (small, 8/8, with the control), and the Molek mistakes list. Do not share: the QA gate details, anything from the allowlist, account names, or ledger counts of evaluators.

7 Not done / next

LIKHA stage-1 behaviour probes: DONE, see `LIKHA-STAGE1-PROBES.md` (likelihood vs shuffles, samples per tag, greedy-continuation copying); still open there: corpus-membership check, folding of samples. A proper variant-effect benchmark with ClinVar-style labels; the PH163 structure probe on more than the one precomputed seed (needs the pod); screenshots in light mode.